

3-Mark Questions & Answers — DBMS Module 5: Query Optimization & Transaction Concurrency Control

Q1. Explain equivalence rules for relational algebra transformations.

1. **Selection Commutativity:** $\sigma_{C1}(\sigma_{C2}(E)) \equiv \sigma_{C2}(\sigma_{C1}(E))$
2. **Join Commutativity:** $E1 \bowtie E2 \equiv E2 \bowtie E1$
3. **Pushing Selections Down:** $\sigma_C(E1 \bowtie E2) \equiv \sigma_{C(E1)} \bowtie E2$ (if predicate C involves only attributes of E1).

Q2. Explain transaction state transition diagram.

```
Active -> Read/Write -> Partially Committed -> Commit Log Flushed -> Committed
    |                               |
    v                               v
Failed -----> Aborted (Rollback complete)
```

Q3. Differentiate between Strict 2PL and Rigorous 2PL.

Feature	Strict 2PL	Rigorous 2PL
Exclusive Locks (X)	Held until Transaction Commit/Abort.	Held until Transaction Commit/Abort.
Shared Locks (S)	Released during Shrinking phase.	Held until Transaction Commit/Abort.
Cascading Rollback	Prevents cascading aborts.	Prevents cascading aborts + strict serial order.

Q4. Explain the Validation-Based (Optimistic) concurrency control protocol.

Optimistic Protocol assumes conflicts are rare and executes in 3 phases:

1. **Read Phase:** Read data and execute updates in private local workspace.
2. **Validation Phase:** Check if workspace updates conflict with committed transactions.
3. **Write Phase:** Copy workspace updates to persistent database if validation succeeds.

Q5. Explain Write-Ahead Logging (WAL) protocol.

The **Write-Ahead Logging (WAL)** protocol dictates that log records describing data modifications **MUST** be written and flushed to non-volatile disk storage **BEFORE** the actual dirty database memory blocks are written to disk.